

JEE(Main + Advanced) : Leader & Enthusiast Course (SCORE-3.0)
ANSWER KEY
PAPER-1
PART-1 : PHYSICS

SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	2.25	648.00	1.14	11.42	72.00	96.00	2.54	2.47	240.59	0.14
SECTION-III	Q.	1	2	3	4	5	6	7	8	9	10
	A.	6	1	8	4	2	2	2	3	2	2

PART-2 : CHEMISTRY

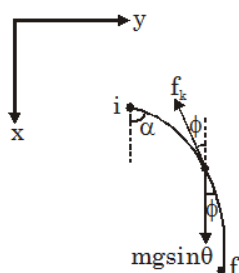
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	16.00	27.00	54.44	2.00	4.00	7.00	10.00	41.50	0.46	2.50
SECTION-III	Q.	1	2	3	4	5	6	7	8	9	10
	A.	5	8	2	6	4	6	6	8	6	5

PART-3 : MATHEMATICS

SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	42.66 or 42.67	2.00	2.57	3.00	36.00	0.70 or 0.71	20.00	3280.00	106.00	3.33 or 3.34
SECTION-III	Q.	1	2	3	4	5	6	7	8	9	10
	A.	6	2	0	1	9	4	1	8	7	6

HINT – SHEET
PART-1 : PHYSICS
SECTION-II
1. Ans (2.25)

f_r must be zero because the block's final speed and hence its final centripetal acceleration must approach zero, which can only happen if the frictional force points up the plane, opposite the weight component down the plane.



$$\frac{dv_x}{dt} = g \sin \theta - \mu_k g \cos \theta \cos \phi \quad \dots(1)$$

$$\frac{dv}{dt} = g \sin \theta \cos \phi - \mu_k g \cos \theta \quad \dots(2)$$

$$\text{Eliminate } \cos \phi. \quad \mu \frac{dv}{dt} = g \cos \theta (\tan^2 \theta - \mu_k^2) - \tan \theta \frac{dv_x}{dt} \quad \dots(3)$$

after simplifying. Now integrate both sides over the time t from the initial to the final point to obtain

$$\mu_k (0 - v_0)$$

$$= g \cos \theta (\tan^2 \theta - \mu_k^2) t - \tan \theta (0 - v_0 \cos \alpha) \quad \dots(4)$$

$$\text{which rearranges as } t = \frac{v_0 (\mu_k + \tan \theta \cos \alpha)}{g \cos \theta (\mu_k^2 - \tan^2 \theta)} \quad \dots(5)$$

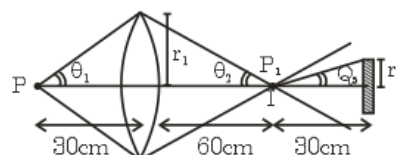
2. Ans (648.00)

$$\tau = F \times 2r$$

$$\alpha = \frac{2Fr}{I}$$

$$\text{Work done} = \tau \theta$$

$$= \tau \times \frac{1}{2} \propto t^2 = 648 \text{ J}$$

3. Ans (1.14)


Power incident on lens

$$\rho_1 = \frac{P}{4\pi} 2\pi (1 - \cos \theta_1)$$

Power incident on Area

$$= \frac{P_1}{2\pi(1 - \cos \theta_2)} 2\pi(1 - \cos \theta_3)$$

$$P_2 = \frac{P}{2} \frac{(1 - \cos \theta_1)(1 - \cos \theta_2)}{(1 - \cos \theta_2)}$$

$$P_2 = \frac{P}{2} \frac{2\sin^2\left(\frac{\theta_1}{2}\right) \cdot 2\sin^2\left(\frac{\theta_3}{2}\right)}{2\sin^2\left(\frac{\theta_2}{2}\right)}$$

$\therefore$ Angle are very small

$$P_2 = P \frac{\theta_1^2 \cdot \theta_3^2}{4Q_2^2} = \frac{P \cdot \left(\frac{r_1}{30}\right)^2 \cdot \left(\frac{r_2}{30}\right)^2}{4\left(\frac{r_1}{60}\right)^2}$$

$$P_2 = \frac{4Pr_2^2}{4 \times (30)^2}$$

$$\pi r_2^2 = 0.5 \Rightarrow r_2^2 = \frac{1}{2\pi}$$

$$P_2 = \frac{2P}{3600\pi}$$

No. of photons incident

$$= \frac{P_2}{(hc/\lambda)} = \frac{P_2 \lambda}{hc}$$

$$N = \frac{2P}{3600\pi} \times \frac{4000}{12420}$$

No. of e^- emitted per unit time

$$n = \frac{20}{100} N$$

$$n = \frac{1}{5} \times \frac{2P \times 4000}{3600\pi \times 12420 \text{ eV}}$$

i = charge emitted per unit time

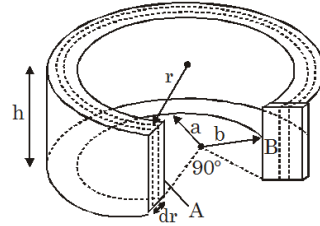
$$i = ne$$

$$i = \frac{1}{5} \times \frac{2P \times 4000e}{3600\pi \times 12420 \text{ eV}}$$

$$i = \frac{2P \times 4000}{5 \times 3600\pi \times 12420} \text{ A}$$

$$i = 1.14 \text{ mA}$$

4. Ans (11.42)



$$\text{Resistance of strip} = \frac{\rho \left(\frac{3\pi r}{2}\right)}{hdr} = \frac{3\pi \rho r}{2hdr}$$

All such strips are in parallel so overall resistance is given as

$$\frac{1}{R} = \int_a^b \frac{2hdr}{3\pi \rho r}$$

$$\frac{1}{R} = \frac{2h}{3\pi \rho} \ln\left(\frac{b}{a}\right)$$

$$R = \frac{3\pi \rho}{2h \ln 2} \text{ (as } b = 2a)$$

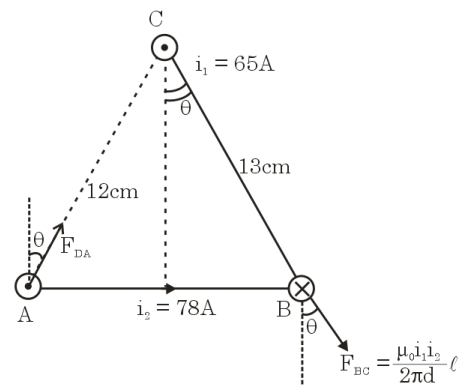
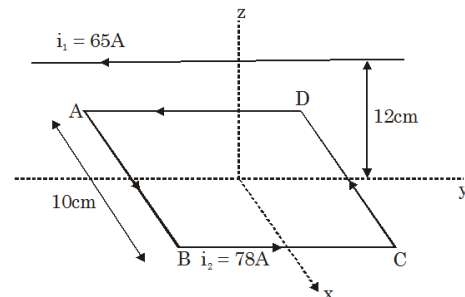
$$= \frac{3}{2} \times \frac{\pi \times 1.6 \times 10^{-8}}{10^{-2} \times 0.693}$$

$$= 11.42 \times 10^{-6}$$

$$= 11.42 \mu\Omega$$

5. Ans (72.00)

Side View



τ_{net} = Torque to force couple of $F \cos \theta$

$$= (F \cos \theta)(10 \text{ cm})$$

$$= \frac{\mu_0 i_1 i_2}{2\pi d} \ell \cos \theta \cdot (10 \text{ cm})$$

Substituting the values we get

$$\tau = 72 \mu\text{N-m}$$

6. **Ans (96.00)**

$$\frac{A_i + A_r}{A_i - A_r} = \frac{3}{2}$$

$$\frac{A_i}{A_r} = 5 \dots (i)$$

$$\text{Now, } A_r^2 + A_t^2 = A_i^2$$

$$A_t^2 = A_i^2 - A_r^2$$

$$A_t = \sqrt{24} A_r$$

$$\text{Thus } \left(\frac{A_t}{A_i} \right)^2 = \frac{24}{25}$$

$$\frac{I_t}{I_i} = \frac{24}{25} \times 100\% = 96\%$$

8. **Ans (2.47)**

$$(S_1 O - t + \mu_1 t) - (S_2 O - t + \mu_2 t) = \Delta x$$

$$\Delta k = (\mu_1 - \mu_2)t$$

$$\Delta k = 0.1 \times 20 \times 10^{-6}$$

$$\Delta \phi = \frac{2\pi}{gl} \Delta x = \frac{2\pi}{4 \times 10^{-7}} \times 2 \times 10^{-6}$$

$$\Delta \phi = 10\pi$$

Max

$$I_{II} = \left(\sqrt{\frac{3}{4}} I_0 + \sqrt{\frac{I_0}{2}} \right)^2 \dots (i)$$

$$I_I = 4I_0 = 4 \text{ (given)}$$

$$\Rightarrow I_0 = 1 \dots (ii)$$

$$I_{II} = \left(\frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \right)^2 I_0 = 2.47$$

9. **Ans (240.59)**

$$\text{Area} = \frac{\pi ab}{2} + b \times ae$$

$$\text{Time taken : } \frac{\frac{\pi ab}{2} + \frac{ab}{2}}{\pi ab} \frac{\pi}{2} + \frac{1}{2} \times 365 = 240.59 \text{ days}$$

10. **Ans (0.14)**

Thickness of slip guage

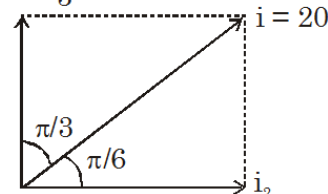
$$= 4.10 - 2.12 = 1.98 \text{ mm}$$

$$\text{Thus, error} = 2.12 - 1.98 = 0.14 \text{ mm}$$

SECTION-III

1. **Ans (6)**

$$i_1 = 20 \cos \frac{\pi}{3}$$



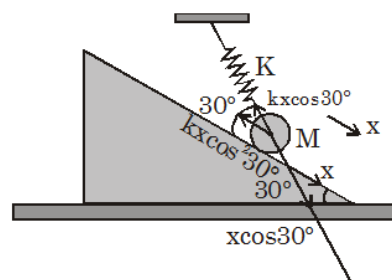
$$i_1 = 10 \sin \left(100\pi t + \frac{\pi}{3} \right)$$

$$V = 220\sqrt{2} \sin(100\pi t)$$

$$Z_1 = \frac{V_{rms}}{i_{rms}} = \frac{220}{\frac{10}{\sqrt{2}}}$$

$$Z_1 = 22\sqrt{2} \Omega$$

2. **Ans (1)**



$$F = -Kx \cos^2 30^\circ$$

$$ma = -(K \cos^2 30^\circ)x$$

$$a = -\left(\frac{K}{m} \frac{3}{4} \right) x$$

$$\omega = \sqrt{\frac{3K}{4m}}$$

$$\omega = \sqrt{\frac{3}{4m} \times \frac{4m}{3}} = 1 \text{ rad/sec}$$

3. **Ans (8)**

$$f_0 = \frac{v}{2\ell} = 200 \text{ Hz} \dots (i)$$

$$f_1 = \frac{v}{2\ell_1} \text{ \& } f_2 = \frac{v}{2\ell_2}$$

$$\text{Beats per sec} = f_1 - f_2 = \frac{v}{2} \left(\frac{1}{\ell_1} - \frac{1}{\ell_2} \right)$$

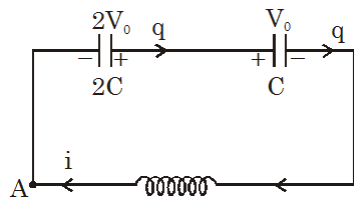
$$y = \frac{v}{2} \left(\frac{\ell_2 - \ell_1}{\ell_1 \ell_2} \right)$$

$$y = \frac{v}{2} \left(\frac{2}{49 \times 51} \right)$$

$$= 200 \times 100 \times \left(\frac{2}{49 \times 51} \right) = 16$$

$$\frac{16}{2} = 8$$

4. Ans (4)



$$\frac{(4CV_0 - q)}{2C} - \frac{CV_0 + q}{C} - L \frac{di}{dt} = 0$$

$$2V_0 - \frac{q}{2C} - V_0 - \frac{q}{C} = L \frac{di}{dt}$$

$$L \frac{d^2q}{dt^2} = V_0 - \frac{3q}{2C}$$

$$\frac{d^2q}{dt^2} = -\frac{3q}{2LC} + \frac{V_0}{L}$$

Solving above equation, we have

$$q = q_0 \sin(\omega t + \delta)$$

where q_0 is maximum value of q and $\omega = \sqrt{\frac{3}{2LC}}$

for maximum q

$$2V_0 - \frac{q}{2C} = V_0 + \frac{q}{C}$$

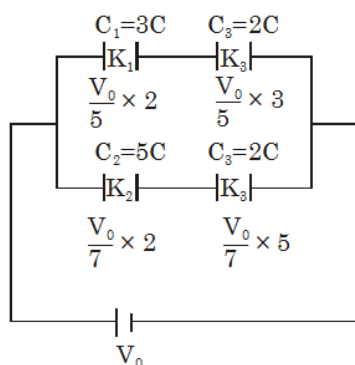
$$q = \frac{2CV_0}{3} = q_0$$

$$\text{Current, } i = \frac{dq}{dt}$$

$$i_{\max} = q_0 \omega$$

$$= \frac{2CV_0}{3} \sqrt{\frac{3}{2LC}} = 4A$$

5. Ans (2)



Ratio of breakdown voltages :

$$V_{B1} : V_{B2} : V_{B3} = 1 : 2 : 3$$

Hence C_1 will breakdown first.

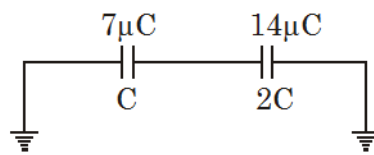
Electric field in C_1 ,

$$E_1 = \frac{\frac{2V_0}{5}}{\frac{d}{2}} = \frac{4V_0}{5d}$$

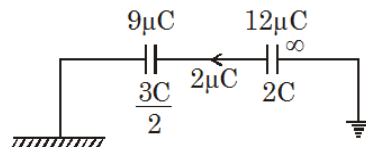
$$= 2 \times 10^3 \text{ V/m} = 2 \text{ V/mm}$$

6. Ans (2)

Initial circuit diagram



Final circuit diagram



$2\mu C$ charge will flow through thin wire

7. Ans (2)

Applying snell's law at each surface

At B, $\sin i = n_1 \sin r_1$

$$\Rightarrow \sin^2 i = n_1^2 \sin^2 r_1$$

At C, $n_1 \sin(90 - r_1) = n_2 \sin r_2$

$$\Rightarrow n_1^2 \cos^2 r_1 = n_2^2 \sin^2 r_2$$

At D, $n_2 \sin(90 - r_2) = n_3 \sin r_3$

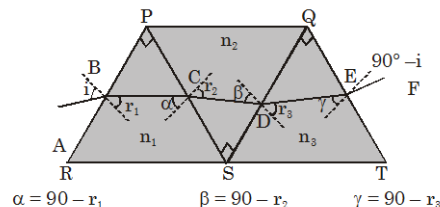
$$\Rightarrow n_2^2 \cos^2 r_2 = n_3^2 \sin^2 r_3$$

At E, $n_3 \sin(90 - r_3) = \sin(90 - i)$

$$\Rightarrow n_3^2 \cos^2 r_3 = \cos^2 i$$

Adding all above equations gives,

$$1 + n_2^2 = n_1^2 + n_3^2$$



8. Ans (3)

$$B = \frac{\mu_0 i r^2}{2(x^2 + r^2)^{3/2}}$$

$$= \frac{\mu_0 i r^2}{2\ell^3} \quad (i = qf)$$

$$= \frac{\mu_0 q r r^2}{2\ell^3}$$

$$= \frac{\mu_0 q f (\ell \sin 60^\circ)^2}{2\ell^3}$$

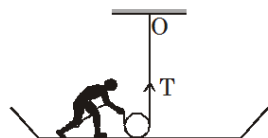
$$= \frac{\mu_0 q \omega \times 3}{16\pi \ell}$$

$$= \frac{4\pi \times 10^{-7} \times 2 \times 10^{-6} \times 5 \times 3}{16\pi \times (\frac{1}{4})}$$

$$= 3 \times 10^{-12} \text{ T}$$

$$= 3 \text{ pT}$$

9. Ans (2)



$$T - (M + m)g = (M + m)a$$

$$a = \frac{50}{75} = \frac{2}{3} \text{ m/s}^2$$

$$W = \Delta k$$

$$W = k_2 - k_1$$

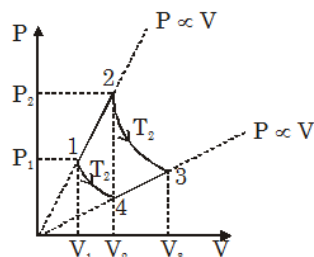
$$= \frac{1}{2}(75)(at)^2$$

$$= \frac{1}{2}(75)\left(\frac{2}{3} \times 2\right)^2$$

$$= \frac{75 \times 8}{9} = \frac{25 \times 8}{3}$$

$$\frac{3W}{100} = \frac{3 \times 25 \times 8}{100 \times 3} = 2$$

10. Ans (2)



$$\frac{P_2}{P_1} = 2$$

$$\frac{V_2}{V_1} = \frac{1}{2}$$

$$\frac{P_1}{P_2} = \frac{1}{2}$$

$$\frac{P_1}{P_2} = \frac{1}{2} \& \frac{V_1}{V_2} = \frac{1}{2}$$

$$P_1 V_1 = P_4 V_2$$

$$P_2 V_2 = P_3 V_3$$

$$\text{Now } \frac{V_3}{V_2} = \frac{P_3}{P_4} = \frac{P_2 V_2}{P_1 V_1} \times \frac{V_2}{V_3}$$

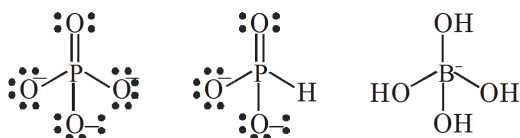
$$\frac{V_3^2}{V_2^2} = \frac{P_2}{P_1} \times \frac{V_2}{V_1}$$

$$= 2 \times 2$$

PART-2 : CHEMISTRY

SECTION-II

2. Ans (27.00)



7. Ans (10.00)



$$95\text{mg} \quad 100\text{mg}$$

$$9.5\text{mg} \quad 10\text{mg}$$

$$H = \frac{10}{10^6} \times 10^6 = 10\text{ppm}$$

8. Ans (41.50)

$$E_a = 10^{-2} \times RT + 2R$$

$$= 10^{-2} \times 8.3 \times 300 + 2 \times 8.3$$

$$= 24.9 + 16.6$$

$$= 41.5$$

9. Ans (0.46)

$$E = (-0.34 + 0.80) - \frac{0.06}{2} \log \frac{[\text{Cu}^{2+}]}{[\text{Ag}^+]^2}$$

$$= 0.46 \text{ Volt}$$

10. Ans (2.50)

$$\frac{P_C \cdot V_{c,m}}{R \cdot T_C} = \frac{3}{8}$$

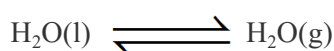
$$\frac{\frac{40}{\pi} \times \left(3 \times 4 \times \frac{4}{3} \pi r^3 \times 10^3 \frac{\text{cm}^3}{\text{mol}} \times 6 \times 10^{23}\right)}{0.08 \times 200} = \frac{3}{8}$$

$$r^3 = \frac{1}{40 \times 4 \times 4} \times 10^{-26} = \frac{1}{4^3} \times 10^{-27}$$

$$r = \frac{1}{4} \times 10^{-9} \text{ cm} = 0.25 \times 10^{-9} \text{ cm} = 2.5 \text{ pm}$$

SECTION-III

2. Ans (8)



$$\Delta_r G^\circ = \Delta_f G^\circ \text{H}_2\text{O(g)} - \Delta_f G^\circ [\text{H}_2\text{O(l)}]$$

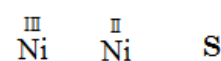
$$= -RT \ln K$$

$$= -RT \ln P_{\text{H}_2\text{O}}$$

$$= -2.303 \times \frac{8.314}{1000} \times 300 \log \frac{30}{750}$$

$$= +8 \text{ kJ (nearest integer)}$$

3. Ans (2)



$$x \quad \left(\frac{3}{4} - x\right) \quad 1.00$$

$$3x + 2 \left(\frac{3}{4} - x\right) = +2$$

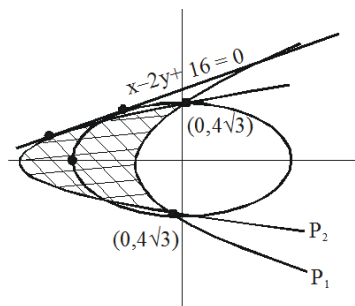
$$x = 2 - 1.5$$

$$x = 0.5$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (42.66 or 42.67)



equation of common tangent is

$$y = \frac{1}{2}x + 8 \quad \begin{cases} m = \frac{1}{2} \\ c = 8 \end{cases}$$

Let equation of $P_1 : y^2 = 4a(x - \alpha)$

and $P_2 : y^2 = 4b(x - \beta)$

both passes through $(0, \pm 4\sqrt{3})$

$$\Rightarrow \alpha = -\frac{12}{a}\beta = -\frac{12}{b}$$

$$\Rightarrow P_1 : y^2 = 4a\left(x + \frac{12}{a}\right)$$

$$\text{and } P_2 : y^2 = 4a\left(x + \frac{12}{b}\right)$$

equation of tangents are

$$y = m\left(x + \frac{12}{a}\right) + \frac{a}{m}, y = m\left(x + \frac{12}{b}\right) + \frac{b}{m}$$

comparing with $y = \frac{1}{2}x + 8$

$$\Rightarrow m = \frac{1}{2}, \frac{6}{a} + 2a = 8, \frac{6}{b} + 2b = 8$$

$$\Rightarrow a = 3, b = 1$$

$$\Rightarrow \alpha = -4, \beta = -12$$

equation of both the parabola are

$$y^2 = 12(x + 4), y^2 = 4(x + 12)$$

$$\begin{aligned} \Rightarrow A &= \int_{-4\sqrt{3}}^{4\sqrt{3}} \left[\left(\frac{y^2}{12} - 4 \right) - \left(\frac{y^2}{4} - 12 \right) \right] dy \\ &= \frac{128}{\sqrt{3}} \end{aligned}$$

2. Ans (2.00)

$$x(f(x)+1) = x^2 + \frac{x}{e} + \int_0^{x^2} f\left(\frac{t}{x}\right) dt \quad \forall x \in \mathbb{R}_0$$

$$\text{Let } \frac{t}{x} = z \Rightarrow \frac{1}{x} dt = dz$$

$$\Rightarrow x(f(x)+1) = x^2 + \frac{x}{e} + x \int_0^x f(z) dz$$

$$\Rightarrow f'(x) = 1 + f(x) \Rightarrow f(1) = 0$$

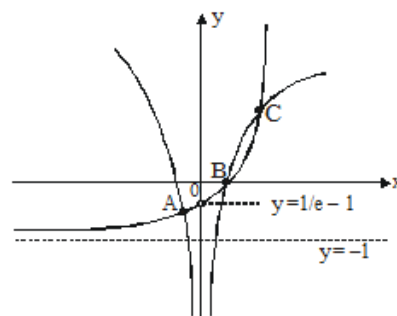
$$\frac{f'(x)}{1+f(x)} = 1 \quad (\because f(x) \neq -1)$$

$$\ln(1+f(x)) = x + c$$

$$\because f(1) = 0 \Rightarrow c = -1$$

$$\Rightarrow f(x) = e^{x-1} - 1$$

Now, $e^{x-1} - 1 = \ln x^2$ intersects at three points as shown in figure.



$$\Rightarrow A(x_1, y_1), B(x_2, y_2), C(x_3, y_3)$$

$$\Rightarrow n = 3, y_1 \in \left(\frac{1}{e} - 1, -1 \right), y_2 = 0$$

$$\Rightarrow [y_1] + [y_2] + n = 2$$

3. Ans (2.57)

$$|(A^{-1} - I)^2| = |A^{-2} - 2A^{-1} + I|$$

$$= \frac{|A^{-1} - 2I + A|}{|A|}$$

$$\because A + A^{-1} = \frac{1}{7} \begin{bmatrix} 12 & 25 & 21 \\ 20 & 23 & 77 \\ -14 & 21 & 21 \end{bmatrix} \text{ and } |A| = -7$$

$$\therefore |(A^{-1} - I)^2| = \frac{\begin{vmatrix} -\frac{2}{7} & \frac{25}{7} & 3 \\ \frac{20}{7} & \frac{9}{7} & 11 \\ -2 & 3 & 1 \end{vmatrix}}{-7} = \frac{324}{49}$$

$$\Rightarrow \text{Its positive square root} = \frac{18}{7}$$

4. Ans (3.00)

$$(yx^2 - e^{y-3})dy = y^2 x dx$$

$$\Rightarrow \frac{dx}{dy} = \frac{yx^2 - e^{y-3}}{y^2 x}$$

$$\Rightarrow x \frac{dx}{dy} = \frac{x^2}{y} - \frac{e^{y-3}}{y^2}$$

$$\Rightarrow x \frac{dx}{dy} - \frac{x^2}{y} = -\frac{e^{y-3}}{y^2}$$

Put $x^2 = t$

$$\Rightarrow 2x \frac{dx}{dy} = \frac{dt}{dy}$$

$$\Rightarrow \frac{1}{2} \frac{dt}{dy} - \frac{t}{y} = -\frac{e^{y-3}}{y^2}$$

$$\Rightarrow \frac{dt}{dy} - \frac{2}{y} \cdot t = \frac{-2}{y^2} \cdot e^{y-3}$$

Integrating factor

$$I.F = \int e^{-\frac{2}{y}} dy = e^{-2 \ln y} = \frac{1}{y^2}$$

$$t \cdot \frac{1}{y^2} = \int \frac{-2}{y^2} \cdot \frac{e^{y-3}}{y^2} dy$$

$$\frac{t}{y^2} = \frac{2}{3} e^{y-3} + C$$

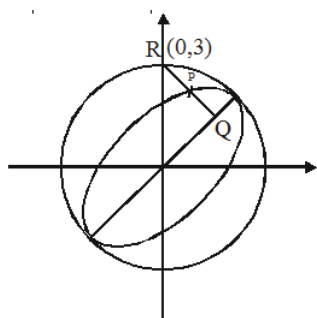
$$\Rightarrow 3t = 2y^2 e^{y-3} + Cy^2$$

$$\Rightarrow \boxed{3x^2 = 2y^2 e^{y-3} + Cy^2}$$

5. Ans (36.00)

Let length of semimajor axis of ellipse = a
and length of semimajor axis of ellipse = b

$$a = 3 \text{ and } RQ = \left| \frac{3.0 - 4.3}{5} \right| = \frac{12}{5}$$



$$\therefore \frac{PQ}{RQ} = \frac{b}{a}$$

$$\Rightarrow \frac{b}{3} = \frac{5}{12}$$

$$\Rightarrow b = \frac{5}{4}$$

Now, distance between foci = $2ae$

$$F = 2\sqrt{a^2 - b^2} = 2\sqrt{9 - \frac{25}{16}}$$

$$\Rightarrow \frac{144F^2}{119} = 36$$

6. Ans (0.70 or 0.71)

$$\vec{p} \cdot \vec{q} = 0, \vec{s} \times \vec{q} = \vec{p} - \vec{s}$$

$$\text{Let } \vec{s} = x\vec{p} + y\vec{q} + z(\vec{p} \times \vec{q})$$

$$(x\vec{p} + y\vec{q} + z(\vec{p} \times \vec{q})) \times \vec{q} = \vec{p} - x\vec{p} - y\vec{q} - z(\vec{p} \times \vec{q})$$

$$x(\vec{p} \times \vec{q}) + z[(\vec{p} \cdot \vec{q})\vec{q} - (\vec{q} \cdot \vec{q})\vec{p}] = (1-x)\vec{p}$$

$$-y\vec{q} - z(\vec{p} \times \vec{q})$$

$$x(\vec{p} \times \vec{q}) - \vec{p}z = (1-x)\vec{p} - \vec{q}y - z(\vec{p} \times \vec{q})$$

comparing $x = -z, y = 0, z = x - 1$

$$\Rightarrow x = \frac{1}{2}, y = 0, z = -\frac{1}{2}$$

$$\Rightarrow \vec{s} = \frac{\vec{p} - \vec{p} \times \vec{q}}{2}$$

$$\Rightarrow |\vec{s} + \vec{p} \times \vec{q}| = \left| \frac{\vec{p} - \vec{p} \times \vec{q}}{2} + \vec{p} \times \vec{q} \right|$$

$$= \frac{|\vec{p} + \vec{p} \times \vec{q}|}{2} = \frac{\sqrt{|\vec{p}|^2 + |\vec{p} \times \vec{q}|^2 + 0}}{2}$$

$$= \frac{1}{\sqrt{2}} = 0.707$$

Alternat

$$\text{Let } \vec{p} = \hat{i}, \vec{q} = \hat{j} \text{ and } \vec{s} = x_1 \hat{i} + x_2 \hat{j} + x_3 \hat{k}$$

Now proceed

7. Ans (20.00)

$$\text{Let } h(x) = px^2 + qx + r \begin{cases} h(0) = -6 = r \\ h'(0) = \beta = q \\ h''(0) = -2\gamma = 2p \end{cases}$$

$$\Rightarrow h(x) = -\gamma x^2 + \beta x - 6$$

$$\therefore h(x) \Rightarrow 4x^4 - \alpha x^3 + \gamma x^2 - \beta x + 6 = 0$$

$$\therefore \text{A.M.} \geq \text{G.M.}$$

$$\Rightarrow \frac{\frac{1}{x_1} + \frac{2}{x_2} + \frac{3}{x_3} + \frac{4}{x_4}}{4} \geq \sqrt[4]{\frac{1.2.3.4}{x_1 x_2 x_3 x_4}}$$

$$\Rightarrow 2 \geq \left(\frac{24}{6/4} \right)^{1/4} \Rightarrow 2 \times 2$$

$$\therefore \text{A.M.} = \text{G.M.}$$

$$\therefore \frac{1}{x_1} = \frac{2}{x_2} = \frac{3}{x_3} = \frac{4}{x_4} = k \text{ (let)}$$

$$\frac{1}{x_1} + \frac{2}{x_2} + \frac{3}{x_3} + \frac{4}{x_4} = 4k = 8 \Rightarrow k = 2$$

$$\therefore x_1 = \frac{1}{2}, x_2 = 1, x_3 = \frac{3}{2}, x_4 = 2$$

$$(2x-1)(x-1)(2x-3)(x-2) = 0$$

$$(2x^2 - 3x + 1)(2x^2 - 7x + 6) = 0$$

8. Ans (3280.00)

$$N = \underline{a} \underline{b} \underline{c} \underline{d} \underline{e}$$

Let x_r be the number of times digit r is used in

$$N, r = 0, 1, 2, 3, \dots, 9.$$

$$\text{If } N \in P \quad x_1 + x_2 + \dots + x_9 = 5 \quad \dots(i)$$

$$\text{If } N \in Q \quad x_0 + x_1 + \dots + x_9 = 5 \quad \dots(ii)$$

$$\text{From (i)} \quad n(P) = {}^{13}C_8$$

$$\text{From (ii)} \quad n(Q) = {}^{14}C_9$$

$$n(P \cap Q) = Q \{ \text{when } a = b = c = d = e \}$$

$$\text{so } n(P \cup Q) = 1287 + 2002 - 9 = 3280$$

9. Ans (106.00)

Chance that a coin is discarded

$$= \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{7}{8}$$

P(atmost two coins remains)

$$\begin{aligned} &= \left(\frac{7}{8}\right)^6 + {}^6C_5 \left(\frac{7}{8}\right)^5 \cdot \frac{1}{8} + {}^6C_4 \left(\frac{7}{8}\right)^4 \cdot \frac{1}{8^2} \\ &= \frac{2401}{8^6} [49 + 42 + 15] = \left(\frac{2401}{8^6}\right) \cdot (106) \end{aligned}$$

10. Ans (3.33 or 3.34)

$$[y] + \{y\}^3 = x, y = f(x)$$

Let $y \in [I-1, I)$, I is an integer

$$(I-1) + (f(x) - (I-1))^3 = x$$

$$f(x) = (I-1) + (x - (I-1))^{1/3}$$

Left hand derivative at $x = I$

$$\begin{aligned} \Rightarrow Lf'(I) &= \lim_{h \rightarrow 0} \frac{((I-1) + (I-h-I+1)^{1/3}) - ((I-1) + 1)}{-h} \\ &= \lim_{h \rightarrow 0} \frac{(1-h)^{1/3} - 1}{-h} \\ &= \frac{1}{3} \\ \Rightarrow \sum_{i=0}^9 \frac{1}{3} &= \frac{10}{3} \end{aligned}$$

SECTION-III

1. Ans (6)

$$x^2 - x + 1 = 0 \begin{cases} \alpha \\ \beta \end{cases}$$

where $\alpha = -\omega$, $\beta = -\omega^2$, ω is non-real cube roots of unity.

$$\begin{aligned} P_{12k+2} &= 2\alpha^{12k+2} + 3\beta^{12k+2} \\ &= 2(-\omega)^{12k+2} + 3(-\omega^2)^{12k+2} \\ &= 2\omega^2 + 3\omega \end{aligned}$$

$$\begin{aligned} P_{6k+1} &= 2\alpha^{6k+1} + 3\beta^{6k+1} \\ &= 2(-\omega)^{6k+1} + 3(-\omega^2)^{6k+1} \\ &= -2\omega - 3\omega^2 \end{aligned}$$

$$\begin{aligned} \Rightarrow P_{12k+2} - P_{6k+1} &= (2\omega^2 + 3\omega) - (-2\omega - 3\omega^2) \\ &= -2 - 3 = -5 \end{aligned}$$

$$\Rightarrow \sum_{k=0}^{6n-1} |P_{12k+2} - P_{6k+1}| = \sum_{k=0}^{6n-1} 5 = 30n$$

$$\text{Now, } P_k = 2\alpha^k + 3\beta^k$$

$$\therefore \alpha^2 - \alpha + 1 = 0$$

$$2\alpha^{k+2} - 2\alpha^{k+1} + 2\alpha^k = 0$$

$$\text{similarly } 3\beta^{k+2} - 3\beta^{k+1} + 3\beta^k = 0$$

$$(2\alpha^{k+2} + 3\beta^{k+2}) - (2\alpha^{k+1} + 3\beta^{k+1}) + (2\alpha^k + 3\beta^k) = 0$$

$$\Rightarrow P_{k+2} + P_k = P_{k+1}$$

$$\Rightarrow \sum_{k=0}^{6n-2} (P_{k+2} + P_k) = \sum_{k=0}^{6n-2} P_{k+1}$$

$$= 2 \sum_{k=0}^{6n-2} \alpha^{k+1} + 3 \sum_{k=0}^{6n-2} \beta^{k+1}$$

$$= \frac{2\alpha \{ \alpha^{6n-1} - 1 \}}{\alpha - 1} + \frac{3 \{ \beta^{6n-1} - 1 \}}{\beta - 1}$$

$$\therefore x(x-1) = -1 \Rightarrow \alpha - 1 = -\frac{1}{\alpha} \text{ and } \beta - 1 = \frac{1}{\beta}$$

$$= -2\alpha^2 \{ \alpha^{6n-1} \} - 3\beta^2 \{ \beta^{6n-1} - 1 \}$$

$$= -2\alpha^{6n+1} + 2\alpha^2 - 3\beta^{6n+1} + 3\beta^2$$

$$= -2\alpha + 2\alpha^2 - 3\beta + 3\beta^2 (\because \alpha^{6n} = \beta^{6n} = 1)$$

$$= 2(-1) + 3(-1) = -5 (\because \alpha^2 - \alpha = \beta^2 - \beta = -1)$$

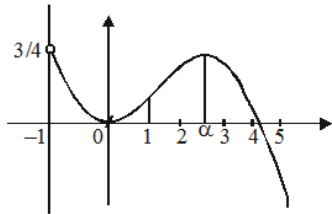
$$\begin{aligned} \Rightarrow \frac{\sum_{k=0}^{6n-1} |P_{12k+2} - P_{6k+1}|}{n \left| \sum_{k=0}^{6n-2} (P_{k+2} + P_k) \right|} &= \frac{30n}{n \cdot 5} = 6 \end{aligned}$$

2. Ans (2)

$$f(x) = (1+x) \ln(1+x) - x - \frac{x^2}{4}$$

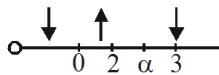
$$\lim_{x \rightarrow -1} f(x) = 0 + 1 - \frac{1}{4} = \frac{3}{4}$$

$$\text{and } f(0) = 0$$



$$f'(x) = \ln(1+x) - \frac{x}{2}$$

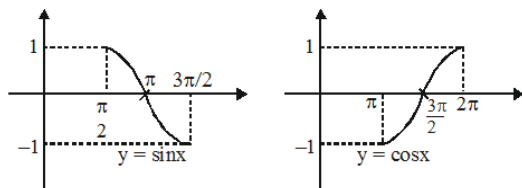
$$f'(0) = 0, f'(2) > 0, f'(3) < 0$$



$$\Rightarrow \alpha \in (2, 3)$$

3. Ans (0)

$$\text{Let } \cos I(x) = A, \cos I, \sin I(x\sqrt{3}) = B$$



$$\because A - B = C \Rightarrow \cos A \cos B + \sin A \sin B = \cos C$$

$$x\sqrt{1-x^2} + \sqrt{1-x^2} \cdot x = x\sqrt{3}$$

$$\Rightarrow x = 0, \frac{1}{2}, -\frac{1}{2}$$

Now, check $x = 0$

$$\cos I(0) - \sin I(0) = \cos I(0)$$

$$\sin I(0) = 0$$

$\pi = 0$ not possible

check at $x = \frac{1}{2}$

$$\cos I\left(\frac{1}{2}\right) - \sin I\left(\frac{1}{2}\right) = \cos I\left(\frac{\sqrt{3}}{2}\right)$$

$$\frac{5\pi}{3} - \frac{5\pi}{6} = \frac{11\pi}{6} \quad (\text{Not possible})$$

$\Rightarrow x = \frac{1}{2}$ is not possible

Now at $x = -\frac{1}{2}$

$$\cos I\left(-\frac{1}{2}\right) - \sin I\left(-\frac{1}{2}\right) = \cos I\left(-\frac{\sqrt{3}}{2}\right)$$

$$\frac{4\pi}{3} - \frac{7\pi}{6} = \frac{7\pi}{6} \quad \text{not possible, so number of}$$

solutions is 0.

4. Ans (1)

$$x + y - z = 1$$

$$(m-1)x + (3m-7)y - 5z = -3$$

$$(m-2)x + (2m-5)y = -4$$

$\therefore$ all the three planes intersect at a line

$$\therefore \begin{vmatrix} 1 & 1 & -1 \\ m-1 & 3m-7 & -5 \\ m-2 & 2m-5 & 0 \end{vmatrix} = 0$$

$$\Rightarrow m = 3, -2$$

$$\text{at } m = 3 \quad x + y - z = 1$$

$$2x + 2y - 5z = -3$$

$$x + y = -4$$

system does not have any solution but at $m = -2$

$$P_1 : x + y - z = 1$$

$$P_2 : 3x + 13y + 5z = 3$$

$$P_3 : 4x + 9y = 4$$

$$\Rightarrow 5P_1 + P_2 = 2P_3$$

$\Rightarrow$ direction ratio of line of intersection are given

by

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & -1 \\ 3 & 13 & 5 \end{vmatrix} = 18\hat{i} - 8\hat{j} + 10\hat{k}$$

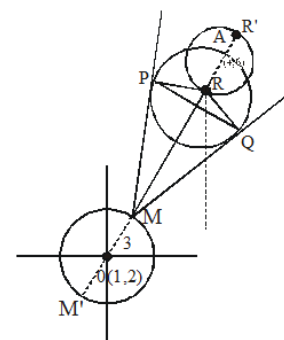
$$\equiv \langle 9, -4, 5 \rangle$$

$$\Rightarrow \alpha = 9\lambda, b = -4\lambda, c = 5\lambda$$

5. Ans (9)

Minimum value of length of diameter is MR

and its maximum value is M'R'



$$\therefore MR = OA - 3 - 1 = 5 - 4 = 1 = \ell$$

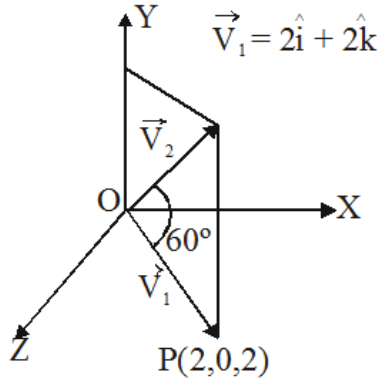
$$M'R' = MR + RR' + MM' = 1 + 2 + 6 = 9 = L$$

$$\Rightarrow \frac{L}{\ell} = 9$$

6. Ans (1)

Equation of plane perpendicular to XOZ plane

is $P : x - z = 0$ $\vec{V}_2$ is coplanar with $\vec{V}_1$ and $\hat{j}$



$$\Rightarrow \vec{V}_2 = x\vec{V}_1 + y\hat{j}$$

$$\Rightarrow \vec{V}_2 = x(2\hat{i} + 2\hat{k}) + y\hat{j}$$

$$\therefore |\vec{V}_2| = |\vec{V}_1| = 2\sqrt{2}$$

$$\therefore 4x^2 + y^2 + 4x^2 = 8 \quad \dots(1)$$

$$\text{Now, } \vec{V}_2 \cdot \vec{V}_1 = |\vec{V}_2| |\vec{V}_1| \cos \frac{\pi}{3} = 4$$

$$4 = (x(2\hat{i} + 2\hat{k}) + y\hat{j}) \cdot (2\hat{i} + 2\hat{k})$$

$$4 = 8x \Rightarrow x = \frac{1}{2} \Rightarrow y = \pm\sqrt{6}$$

$$\Rightarrow \vec{V}_2 = \hat{i} + \sqrt{6}\hat{j} + \hat{k} \quad (\because \vec{V}_2 \cdot \hat{j} > 0)$$

$$\text{Now, } \vec{V}_3 = \vec{V}_1 + x\hat{i}$$

$$\therefore \vec{V}_3 \cdot \vec{V}_1 = 0 \Rightarrow 8 + 2x = 0 \Rightarrow x = -4$$

$$\Rightarrow \vec{V}_3 = 2\hat{i} + 2\hat{k} - 4\hat{i}$$

$$\vec{V}_3 = -2\hat{i} + 2\hat{k}$$

$\Rightarrow$ volume of tetrahedron

$$V = \begin{vmatrix} 2 & 0 & 2 \\ \frac{1}{6} & 1 & \sqrt{6} \\ -2 & 0 & 2 \end{vmatrix}$$

$$\Rightarrow V = \frac{1}{6} 8\sqrt{6} \Rightarrow \frac{3V}{\sqrt{6}} = 4$$

7. Ans (1)

$$f(x) = \sin^2 x e^{\cot x} \quad x \in (0, \pi)$$

$$f'(x) = \sin^2 x \cdot e^{\cot x} (-\cot^2 x) + e^{\cot x} \cdot \sin 2x$$

$$= e^{\cot x} [\sin 2x - 1] = 0$$

$$\Rightarrow x = \frac{\pi}{4}$$

Now

$$f''(x) = e^{\cot x} \cdot \cos 2x \cdot 2 + (\sin 2x - 1) e^{\cot x} (-\cot^2 x)$$

$$= e^{\cot x} \left[2 \cos 2x + \frac{1 - \sin 2x}{\sin^2 x} \right]$$

$$= e^{\cot x} \left[2 \left(\frac{\cot^2 x - 1}{\cot^2 x + 1} \right) + (1 + \cot^2 x) \left(1 - \frac{2 \cot x}{1 + \cot^2 x} \right) \right]$$

Let $\cot x = t$

$$\Rightarrow 2t^2 - 2 + (1 + t^2)(1 + t^2 - 2t) = 0$$

$$(t - 1)(2t + 2) + (1 + t^2)(t - 1)^2 = 0$$

$$(t - 1)[t^3 - t^2 + 3t + 1] = 0 \quad t \in \mathbb{R}$$

$$\Rightarrow t = 1 \text{ or } t^3 - t^2 + 3t + 1 = 0$$

$$\Rightarrow x = \frac{\pi}{4}$$

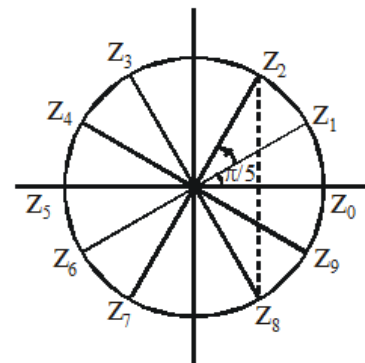
$$\Rightarrow \text{at } x = \frac{\pi}{4} f' \left(\frac{\pi}{4} \right) = f'' \left(\frac{\pi}{4} \right) = 0$$

$\Rightarrow$ Number of value of α is 1

8. Ans (8)

$$z = \cos \frac{\pi}{5} + 2i \sin \frac{\pi}{10} \cos \frac{\pi}{10}$$

$$\Rightarrow z = \cos \frac{\pi}{5} + i \sin \frac{\pi}{5} = e^{i\frac{\pi}{5}}$$



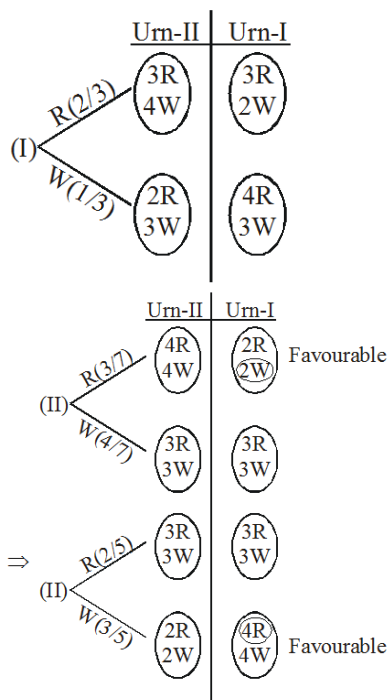
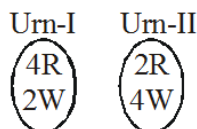
$$\therefore z_p = z_p = e^{ip\frac{\pi}{5}}, p = 0, 1, 2, 3, \dots, 9$$

$$(z_2, z_0) (z_3, z_1) (z_4, z_2) (z_5, z_3) (z_6, z_4) (z_7, z_5) (z_8, z_6)$$

$$(z_9, z_7)$$

$\Rightarrow$ 8 ordered pairs

9. Ans (7)



$\Rightarrow$ Required probability P is given by

$$P = \frac{\frac{1}{3} \times \frac{3}{5}}{\frac{1}{3} \times \frac{3}{5} + \frac{2}{3} \times \frac{3}{7}} = \frac{\frac{1}{5}}{\frac{1}{5} + \frac{2}{7}} = \frac{2 \times 5}{7 + 10} = \frac{7}{17}$$

10. Ans (6)

$$T_r = \frac{4 \cdot (2r-1)^4 + 1}{4(2r)^4 + 1} = \frac{(2(2r-1)^2 + 1)^2 - 4(2r-1)^2}{(2(2r)^2 + 1)^2 - 4(2r)^2}$$

$$= \frac{(2 \cdot (2r-1)^2 - 2 \cdot (2r-1) + 1)(2 \cdot (2r-1)^2 + 2(2r-1) + 1)}{(2(2r)^2 - 2(2r) + 1)(2(2r)^2 + 2(2r) + 1)}$$

$$= \frac{(8r^2 - 8r + 2 - 4r + 2 + 1)(8r^2 - 8r + 2 + 4r - 2 + 1)}{(8r^2 - 4r + 1)(8r^2 + 4r + 1)}$$

$$= \frac{8r^2 - 12r + 5}{8r^2 + 4r + 1}$$

$$P = T_1 \cdot T_2 \cdot \dots \cdot T_n = \frac{1}{13} \cdot \frac{13}{41} \cdot \frac{41}{85}$$

$$\dots \cdot \frac{8n^2 - 12n + 5}{8n^2 + 4n + 1} = \frac{1}{8n^2 + 4n + 1}$$

$$\Rightarrow L = \lim_{n \rightarrow \infty} \frac{n^k}{8n^2 + 4n + 1} = \text{finite}$$

$$\Rightarrow k = 2, L = \frac{1}{8}$$

JEE(Main + Advanced) : Leader & Enthusiast Course (PHASE : SCORE-3.0)
ANSWER KEY
PAPER-2
PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	C	B	C	A	D	A	C	A	A
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	A	C	D	B	B	A	D	C	A	B
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	0.25	0.75	0.50	6.00	1.00	1.66 to 1.67	15.00	8.00	2.00	0.16 to 0.17

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	A	D	D	D	C	D	D	A	A
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	B	B	C	B	B	A	B	D	C	A
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	-0.30	15.00	30.00	4.32	1.00	7.00	4.00	400.00	13.00	6.00

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	D	D	B	A	C	B	A	B	B
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	D	B	A	C	B	B	B	A	B	D
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	0.50	1.5	0.00	17.00	10.00	8.00	35.00	3.00	4.00	2.00

HINT – SHEET
PART-1 : PHYSICS

1. Ans (B)

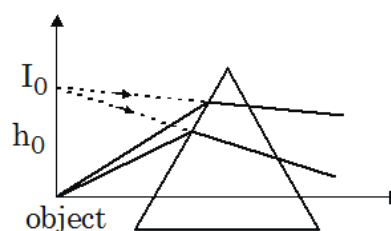
Across PN junction

Resistance ↓ on increasing

Voltage across it hence current increases more

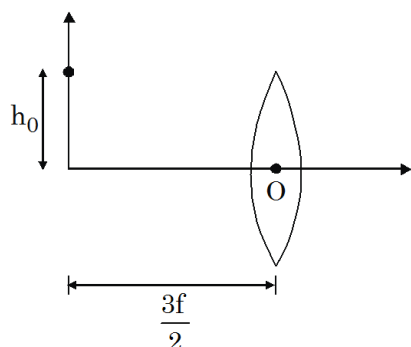
than that of across R.

2. Ans (C)



$$\delta = (n - 1) \times \alpha$$

$$h_0 = f(n - 1) \alpha$$



$$\frac{1}{v} - \frac{2}{(-3f)} = \frac{1}{f}$$

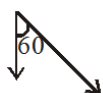
$$\frac{1}{v} = \frac{1}{f} - \frac{2}{3f}$$

$$\frac{1}{v} = \frac{3-2}{3f} \Rightarrow v = 3f$$

$$\frac{h_2}{h_1} = \frac{v}{u} \Rightarrow h_2 = h_0 \frac{3f}{-\frac{3f}{2}}$$

$$h_2 = -2f(n-1)\alpha$$

4. **Ans (C)**



$$\mu \cos 60$$

$$\text{net } I = 2\mu \cos 60 = \mu$$

5. **Ans (A)**

$$E = e\sigma AT^4$$

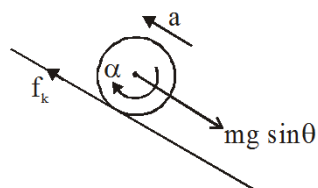
$$\Rightarrow 100 = 0.3 \times 5.62 \times 10^{-8} \times A \times (2780)^4$$

$\therefore$ Surface area of bulb

$$= \frac{100}{101.46 \times 10^4} = 0.98 \times 10^{-4} \text{ m}^2$$

Ans (D)

6.



$$f_k - mg \sin \theta = ma$$

$$f_k R = I\alpha = \frac{mR^2}{2}a$$

$$f_k = \frac{mRa}{2}$$

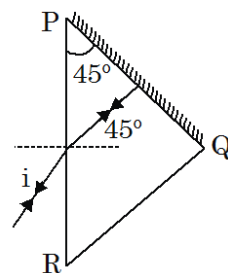
$$Ra = \frac{2f_k}{m} = \frac{2(\mu N)}{m}$$

$$= \frac{2(\mu) mg \cos \theta}{m} (\because \mu = \tan \theta)$$

$$= 2(\tan \theta) g \cos \theta = 2g \sin \theta$$

There is rolling & slipping so, $a \neq R\alpha$
& $a = 0$

7. **Ans (A)**



$$1 \times \sin i = n \sin r$$

$$\sin i = n \sin 45^\circ$$

$$\sin i = \frac{n}{\sqrt{2}}$$

$$\text{For } i = 45^\circ : \frac{1}{\sqrt{2}} = \frac{n}{\sqrt{2}}$$

$$n = 1$$

For $i = 90^\circ$:

$$1 = \frac{n}{\sqrt{2}} \Rightarrow n = \sqrt{2}$$

$$\therefore \text{Range is } \sqrt{2} > n > 1$$

8. **Ans (C)**

The Bragg angle is half the angle through which the beam is deviated. Thus, in this case, the bragg angle

$$\theta = 1/2 \times 60^\circ = 30^\circ$$

by bragg equation, $2d \sin \theta = n\lambda$

Here, $d = 3 \times 10^{-10} \text{ m}$, $\theta = 30^\circ$; $n = 1$, $\lambda = ?$

$$\lambda = \frac{2d \sin \theta}{n} = \frac{2 \times (3 \times 10^{-10}) \sin 30^\circ}{1} \text{ m}$$

$$= 3 \times 10^{-10} \text{ m} = 3 \text{ \AA}$$

9. **Ans (A)**

$$f = \frac{1}{2\pi\sqrt{LC}} \Rightarrow f \propto \frac{1}{\sqrt{C}}$$

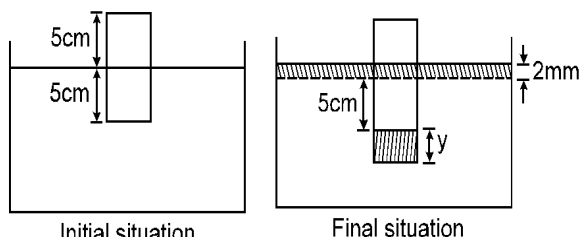
$$f' \propto \frac{1}{\sqrt{16C}} \Rightarrow f' = \frac{f}{4}$$

10. **Ans (A)**

Forward thrust does positive work on the spacecraft causing it to rise higher and its potential energy will increase. With increase in r , the kinetic energy decreases.

11. **Ans (A)**

Let the cube dips further by y cm and water level rises by 2 mm.



Volume of water raised

= volume of extra depth of wood

$$\Rightarrow 100y = (1500 - 100) \frac{2}{10}$$

$$= 1400 \times \frac{2}{10} = 280$$

$$\therefore y = 2.8 \text{ cm}$$

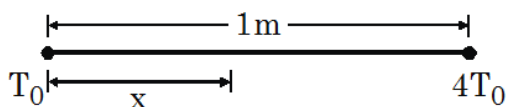
$\therefore$ Extra upthrust

$$\rho_{\text{water}} \times (2.8 + 0.2) \times 100 \text{ g} = mg$$

$$\Rightarrow m = 300 \text{ gm.}$$

$$m = 300 \text{ gm.}$$

12. **Ans (C)**



$$T = T_0 + \frac{4T_0 - T_0}{1} \times x \Rightarrow T = T_0 + 3T_0x$$

$$V = k\sqrt{T} \Rightarrow \frac{dx}{dt} = kT_0^{1/2}(1 + 3x)^{1/2}$$

$$\int_{x=0}^{x=1\text{m}} \frac{dx}{(1 + 3x)^{1/2}} = k\sqrt{T_0} \int_0^t dt$$

$$\frac{2}{3} [\sqrt{1 + 3x}]_0^1 = k\sqrt{T_0} t$$

$$\frac{2}{3} [2 - 1] = k\sqrt{T_0} t \Rightarrow t = \frac{2}{3k\sqrt{T_0}}$$

13. **Ans (D)**

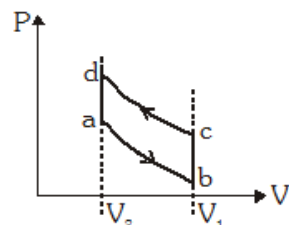
$$P = \frac{\rho}{M_0} RT \Rightarrow \frac{P}{\rho} = \frac{R}{M_0} T$$

Slope of the curve $\propto$ Temperature

Hence cd and ab are isothermal processes.

$$\rho \propto \frac{1}{V}$$

i.e. bc and da are constant volume process



Equivalent PV diagram

(1) and (2) are true.

Temp. in cd process is greater than ab.

Net work done by the gas in the cycle is negative, as is clear by the PV-diagram.

14. **Ans (B)**

Nuclear mass = atomic mass – mass of the atom's electron

$$\therefore N_{\text{nucleus}} = M_{\text{atom}} - Zm_e$$

$$= 6.015125 - 3 \times 0.0005499$$

$$= 6.013478 \text{ U}$$

15. **Ans (B)**

$$t = \frac{n\lambda}{(\mu - 1)}$$

$$\Rightarrow t = \frac{5 \times 6 \times 10^{-5}}{(1.5 - 1)} = 6 \times 10^{-4} \text{ cm}$$

16. **Ans (A)**

$$n \times \frac{4\pi}{3} r^3 = \frac{4\pi}{3} R^3 \Rightarrow R = (n)^{1/3} r$$

$$V = K \frac{nq}{R} \Rightarrow V = K \frac{nq}{(n)^{1/3} r} \Rightarrow V = n^{2/3} \frac{Kq}{r}$$

$$V = (8)^{2/3} = \frac{9 \times 10^9 \times 66 \times 10^{-15}}{1 \times 10^{-3}} = 2.4 \text{ V}$$

17. **Ans (D)**

Here $f = 0.5 \text{ Hz}$; $N = 100$, $A = 0.1 \text{ m}^2$ and $B = 0.01 \text{ T}$. Employing eq.

$$e_0 = NBA (2\pi v)$$

$$= 100 \times 0.01 \times 0.1 \times 2 \times 3.14 \times 0.5$$

$$= 0.314 \text{ V}$$

The maximum voltage is 0.314 V

We urge you to explore such alternative possibilities for power generation.

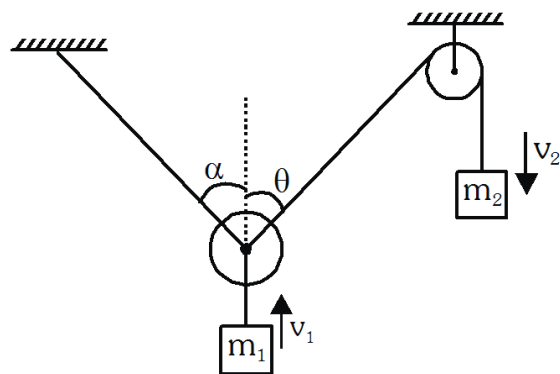
18. **Ans (C)**

By constraint :

$$-v_1 \cos \alpha - v_1 \cos \theta + v_2 = 0$$

$$v_2 = v_1 \cos \theta + v_1 \cos \alpha$$

$$v_2 > v_1 \cos \theta$$



20. **Ans (B)**

$$\text{Work done} = \int_{v_1}^{v_2} P dv$$

$$\text{But } PV = RT$$

$$P = \frac{RT}{V}$$

$$w = RT \int_{v_1}^{v_2} \frac{dv}{V} = RT \log_e \frac{V_2}{V_1}$$

$$\text{Here } \frac{V_2}{V_1} = 4$$

$$\Delta w = RT \times 2.3026 \log_{10} (4)$$

Here, Δw and R are in the units of work.

$$\text{gain in entropy} = \frac{\Delta H}{T}$$

$$\frac{\Delta W}{JT} = \frac{RT \times 2.3026 \log_{10} 4}{JT}$$

$$= 1.387 \frac{R}{J} \text{ cal/K}$$

PART-1 : PHYSICS

1. **Ans (0.25)**

$$U_i = \frac{KQ^2}{2R} \quad U_f = \frac{KQ^2}{2(2R)}$$

(whole charge will get transformed to outer sphere)

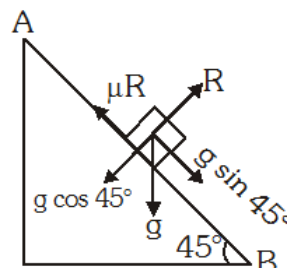
$$\text{so heat} = \frac{KQ^2}{2(2R)} = \frac{KQ^2}{4R}$$

2. **Ans (0.75)**

The acceleration produced in the block the smooth 45° inclined plane, $a_1 = g \sin 45^\circ = g/\sqrt{2}$ and for the rough 45° inclined plane

$$a_2 = g \sin 45^\circ - \mu_k g \cos 45^\circ = \frac{g}{\sqrt{2}} (1 - \mu_k)$$

where μ_k is the coefficient of kinetic friction.



Now time taken to reach the block from A to B down the smooth inclined plane is t_1 and that for the rough inclined plane is t_2 .

$$AB = 0 + \frac{1}{2} a_1 t_1^2 = \frac{1}{2} a_2 t_2^2$$

$$\therefore a_1 t_1^2 = a_2 t_2^2$$

or

$$\frac{t_1}{t_2} = \sqrt{\frac{a_2}{a_1}} = \sqrt{\frac{g/\sqrt{2}(1 - \mu_k)}{g/\sqrt{2}}} = \sqrt{1 - \mu_k}$$

$$\text{Given, } t_2 = 2t_1 \text{ or } \frac{t_1}{t_2} = \frac{1}{2}$$

$$\therefore 1 - \mu_k = \frac{1}{4}$$

$$\text{or } \mu_k = 1 - \frac{1}{4} = 0.75$$

3. **Ans (0.50)**

Potential gradient in first case is

$$K_1 = \frac{\epsilon_0}{\ell}$$

$$\therefore \epsilon = K_1 \times \frac{\ell}{3} \Rightarrow \epsilon = \frac{\epsilon_0}{\ell} \times \frac{\ell}{3} = \frac{\epsilon_0}{3}$$

In the second case

$$\text{Potential gradient } K_2 = \frac{\epsilon_0}{\frac{3\ell}{2}} = \frac{2\epsilon_0}{3\ell}$$

$$\epsilon = K_2 \ell' \Rightarrow \frac{\epsilon_0}{3} = \frac{2\epsilon_0}{3\ell} \ell' \Rightarrow \ell' = \frac{\ell}{2}$$

4. **Ans (6.00)**

$$m \propto v^a p^b g^c$$

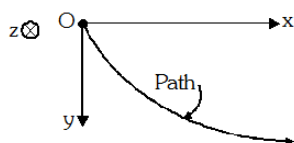
$$[M^1 L^0 T^0] \propto [L^1 T^{-1}]^a [M^1 L^{-3}]^b [L^1 T^{-2}]^c$$

$$b = 1, c = -3, a = 6$$

$$m \propto v^6$$

5. **Ans (1.00)**

Let ball be dropped from origin at $t = 0$.



$\vec{B} = B\hat{k}$; $\vec{g} = g\hat{j}$ At $t = 0$, $\vec{v} = 0$, force due to $\vec{B}$ is in xy plane always, force due to gravity is along Y.

$\therefore$ Ball always moves in XY plane

$\therefore \vec{v} = v_x \hat{i} + v_y \hat{j}$ (at any time)

$$\vec{F} = mg \hat{j} + q(v_x \hat{i} + v_y \hat{j}) \times (B\hat{k})$$

$$\vec{F} = qBv_y \hat{i} + (mg - qBv_x) \hat{j}$$

$\therefore$ Components of acceleration : $a_x = \frac{qBv_y}{m}$

$$(1) \quad a_y = g - \frac{qBv_x}{m} \quad \dots(2) \quad \frac{d}{dt} \text{ of } (2) :$$

$$\ddot{v}_y = -qBv_x \quad \ddot{v}_y = \frac{-qB}{m} \left(\frac{qBv_y}{m} \right) \text{ from}$$

(1)

$$\ddot{v}_y = - \left(\frac{qB}{m} \right)^2 v_y$$

$$v_y = A \sin \left(\frac{qBt}{m} + \phi \right) \quad \dots(3)$$

$$a_y = \frac{AqB}{m} \cos \left(\frac{qBt}{m} + \phi \right) \quad \dots(4) \quad \text{At } t =$$

0, $v_y = 0$ and $a_y = g$

$\therefore$ From (3); $\sin \phi = 0$, & $g = \frac{AqB}{m} \cos \phi$

$\therefore \phi = 0$ and $A = \frac{mg}{Bq} \quad \therefore$

$$v_y = \frac{mg}{Bq} \sin \left(\frac{qBt}{m} \right) \quad \dots(5)$$

6. **Ans (1.66 to 1.67)**

$$q_1 = CE$$

$$q_2 = CE \times \frac{3}{5}$$

7. **Ans (15.00)**

$$\theta = \omega_1 t$$

$$\theta + \pi = \omega_2 t$$

$$\pi = (\omega_2 - \omega_1)t$$

$$t = \frac{\pi}{\omega_2 - \omega_1}$$

$$\& \omega = \sqrt{\frac{GM}{r^3}}$$

8. **Ans (8.00)**

$$I = \frac{34 - 10}{12} = 2 \text{ amp.}$$

$$\Rightarrow V_D = 0$$

$$\therefore V_E = -4 \times 2 = -8 \text{ volt}$$

9. **Ans (2.00)**

$$kx = mg$$

$$\Rightarrow \frac{2mg}{a} x = mg$$

$$\Rightarrow x = \frac{a}{2} \text{ so work done by earth}$$

$$= mgx = mg \frac{a}{2}$$

10. **Ans (0.16 to 0.17)**

$$\frac{hc}{\lambda} = 5 eV_0 + \phi$$

$$\frac{hc}{3\lambda} = eV_0 + \phi \Rightarrow \frac{2hc}{3\lambda} = 4eV_0 \Rightarrow \phi = \frac{hc}{6\lambda}$$

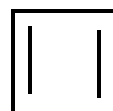
PART-2 : CHEMISTRY

1. **Ans (B)**

Decomposition is an endothermic process and

$$\Delta ng > 0.$$

2. **Ans (A)**

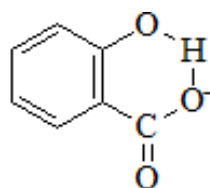


has clyclic conjugation of $4\pi e^-$.

4. **Ans (D)**

Option (D) is correct because

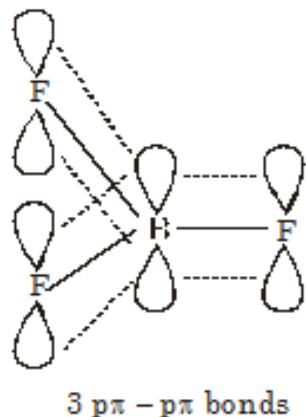
o - hydroxybenzoate ion is stabilized by intramolecular hydrogen bonding.



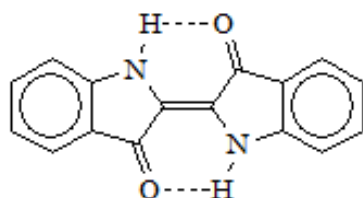
Intramolecular hydrogen bonding

5. **Ans (D)**

$p\pi-d\pi$ back bonding possible when an empty orbital present in the outermost orbit of centre atom. In BF_3 there is maximum possibility of $p\pi-p\pi$ back bonding.



6. **Ans (C)**



⇒ Can show GI

⇒ Trans is more stable than cis due to H-bond.

⇒ It is planer.

7. **Ans (D)**

An organometallic compound contains a covalent bond between metal and carbon atom

$CH_3-Mg-Br$ Grignard's reagent



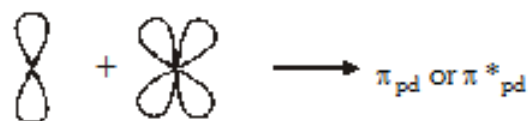
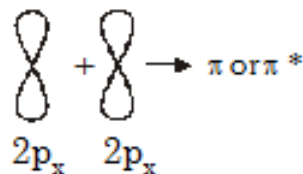
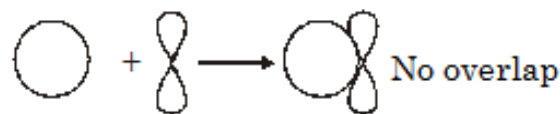
CH_3-Li Methyl lithium

8. **Ans (D)**

$[Ni(NH_3)_6]^{2+}$ is outer orbital complex $\therefore$ 4s, 4p and 4d orbitals (sp^3d^2) hybridisation are involved whereas (A), (B) and (C) are liner complexes ie. d^2sp^3 hybridisation.

9. **Ans (A)**

$1s + 2p_x \rightarrow$ zero overlap



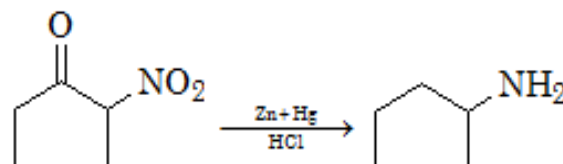
10. **Ans (A)**

Density of ice < Density of water.

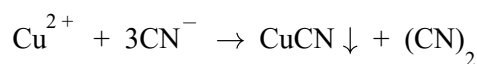
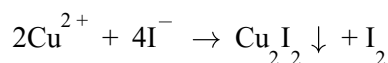
11. **Ans (B)**

3° halide under solvolysis conditions and at high temperature gives elimination product as major product.

12. **Ans (B)**



13. **Ans (C)**



14. **Ans (B)**

Lyophilic colloids are reversible sols.

15. **Ans (B)**

At low temperature ($-70^\circ C$); $LiAlH_4$ only reduces carbonyl group.

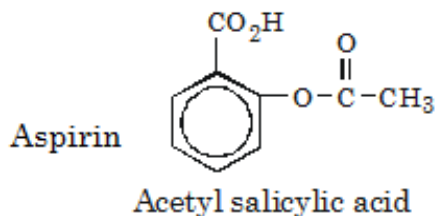
16. **Ans (A)**

Reaction through cyclic halonium ion.

17. **Ans (B)**

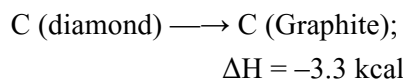
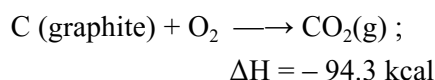
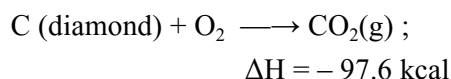
$\text{NaCl} + \text{AgNO}_3 \rightarrow \text{AgCl} \downarrow (\text{White}) + \text{NaNO}_3$
 $;$ $\text{AgS} \downarrow (\text{black})$; Ag_2CO_3 and Ag_2SO_3 dissolves
 in dil. HNO_3 liberating CO_2 and SO_2
 respectively.

20. **Ans (A)**



PART-2 : CHEMISTRY

1. **Ans (-0.30)**



Heat required to convert 12 gram diamond to graphite = 3.3

$\therefore$ Heat required to convert 1 gm diamond to graphite = $\frac{3.3}{12} = 0.275$

2. **Ans (15.00)**

Amount of A left in n_1 halves life = $\left[\frac{1}{2}\right]^{n_1} [A]_0$

Amount of B left in n_2 halves life $\left[\frac{1}{2}\right]^{n_2} [B]_0$

At the end

$$\frac{[A]_0}{2^{n_1}} = \frac{[B]_0}{2^{n_2}} = \frac{4}{2^{n_2}} = \frac{4}{2^{n_1}} = \frac{1}{2^{n_2}}$$

$$\frac{2^{n_1}}{2^{n_2}} = 4 \Rightarrow 2^{n_1-n_2} = (2)^2 \therefore n_1 - n_2 = 2$$

$$n_2 = (n_1 - 2)$$

$$\frac{n_1(t_{1/2})_1}{n_2(t_{1/2})_2} = 1 \Rightarrow \frac{n_1 \times 5}{n_2 \times 15} = 1 \Rightarrow \frac{n_1}{n_2} = 3$$

$$n_1 = 3n_2$$

$$n_2 = 3n_2 - 2$$

$$2n_2 = 2$$

$$n_2 = 1$$

$$n_1 = 3$$

$$t = 3 \times 5 = 15 \text{ minutes}$$

3. **Ans (30.00)**

Redox titration

Eq. of $\text{K}_2\text{C}_2\text{O}_4 \cdot 3\text{H}_2\text{C}_2\text{O}_4 \cdot 4\text{H}_2\text{O}$ = Eq. of KMnO_4

$$\frac{0.1 \times V}{1000} = \frac{20 \times 0.05 \times 5}{1000} \Rightarrow V = 50 \text{ ml}$$

n factor of $\text{K}_2\text{C}_2\text{O}_4 \cdot 3\text{H}_2\text{C}_2\text{O}_4 \cdot 4\text{H}_2\text{O}$

for redox titration = 8

for acid base titration = 6

$\therefore$ for acid base titration normality of

$$\text{K}_2\text{C}_2\text{O}_4 \cdot 3\text{H}_2\text{C}_2\text{O}_4 \cdot 4\text{H}_2\text{O} = \frac{0.1}{8} \times 6N$$

Eq. of acid = Eq. of base

$$\frac{0.1 \times 50}{8 \times 1000} = \frac{1}{8} \times \frac{V \text{ ml}}{1000}$$

$$V \text{ ml} = 30 \text{ ml}$$

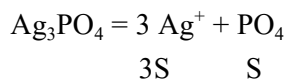
4. **Ans (4.32)**

$$\lambda = K \times \frac{1000}{N}$$

$$1.50 \times 10^{-4} \times 10^4 = 9 \times 10^{-5} \times 10^{-2} \times \frac{1000}{N}$$

$$N = 6 \times 10^{-5}$$

$$S = M = \frac{N}{n_f} = \frac{6 \times 10^{-5}}{3} = 2 \times 10^{-5} \text{ mol / L}$$



$$K_{sp} = (3S)^3 \cdot S = 27 \cdot S^4 = 27 \times (2 \times 10^{-5})^4$$

$$= 4.32 \times 10^{-18}$$

5. **Ans (1.00)**

Ag is correct because impurity of Pb is present in silver can be removed as volatile oxide

(PbO) in the process called cupellation. Impure

Ag is heated in a cupel (boat shaped) shallow

crucible made of bone ash. (A), (C), (D) are not possible.

6. **Ans (7.00)**

$$\Psi(r) = K \cdot e^{-K_1 r} [r^2(r^2 + K_1 r + K_2)]$$

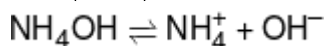
$$\Psi(r) = K \cdot e^{-K_1 r} r^2(r^2 + K_1 r + K_2)$$

Radial nodes = 2

$$n - \ell - 1 = 2$$

$$\text{For } \ell = 2$$

7. **Ans (4.00)**



$$1 \quad 0 \quad 0$$

$$1 - \alpha \quad \alpha \quad \alpha$$

$$c(1 - \alpha) \quad c\alpha \quad c\alpha$$

$$[\text{OH}^-] = c\alpha$$

$$\Rightarrow \frac{50}{100} \times 0.1 = 0.05 \text{ M}$$

$$K_{sp} \text{ of } \text{Zn}(\text{OH})_2 = [\text{Zn}^{2+}] [\text{OH}^-]^2$$

$[\text{OH}^-]$ being common ion, solubility of

$\text{Zn}(\text{OH})_2$ will decrease.

$$\Rightarrow 1 \times 10^{-14} = [\text{Zn}^{2+}] [0.05]^2$$

$$[\text{Zn}^{2+}] = 4 \times 10^{-12} \text{ M.}$$

$$\text{Solubility of } \text{Zn}(\text{OH})_2 = [\text{Zn}^{2+}] = 4 \times 10^{-12} \text{ M}$$

8. **Ans (400.00)**

$\therefore$ 8 g sulphur present in 100 g of compound

$\therefore$ 1 g sulphur present in $\frac{100}{8}$ g of compound

$\therefore$ 32 g sulphur present in $\frac{100 \times 32}{8} = 400 \text{ g}$

Least molecular mass means at least one atom of sulphur must be present.

PART-3 : MATHEMATICS

1. **Ans (A)**

$$\frac{AB}{AC} = \frac{BD}{CD} = \frac{25}{39} \Rightarrow D \equiv \left(\frac{175}{8}, 0 \right)$$

$$\text{Equation of AD} \equiv y = -8 \left(x - \frac{175}{8} \right)$$

$$8x + y = 175$$

2. **Ans (D)**

$$f(x) = \frac{\cos x - \sin x}{1 + (\sin x + \cos x)^2}$$

$f(x)$ is monotonic increasing when $f'(x) > 0$

$$\Rightarrow \frac{\cos x - \sin x}{1 + (\sin x + \cos x)^2} > 0 \Rightarrow \cos x - \sin x > 0$$

$$0$$

$$\Rightarrow \sqrt{2} \cos \left(x + \frac{\pi}{4} \right) > 0$$

$$\Rightarrow -\pi/2 < x + \pi/4 < \pi/2$$

($\because \cos x$ is positive when $-\pi/2 < x < \pi/2$)

$$\therefore -3\pi/4 < x < \pi/4$$

3. **Ans (D)**

$$\begin{bmatrix} \vec{a} & \vec{b} & \vec{c} \end{bmatrix} = \begin{vmatrix} 1 & -1 & 0 \\ x & 1 & 1-x \\ y & x & 1+x-y \end{vmatrix}$$

$$= 1 + x - y - x + x^2 + x + x^2 - xy - y + xy$$

$$= 1 + x - 2y + 2x^2$$

4. **Ans (B)**

Normal vector for the plane containing points $(1 +$

$2\lambda, 2 + 3\lambda, 3 + 4\lambda)$ and

$(2 + 3\mu, 3 + 4\mu, 4 + 5\mu)$ is

$$(2\hat{i} + 3\hat{j} + 4\hat{k}) \times (3\hat{i} + 4\hat{j} + 5\hat{k}) = -\hat{i} + 2\hat{j} - \hat{k}$$

So, equation of plane will be $x - 2y + z = p$

Since, it passes through $(1, 2, 3)$, $p = 0$

Hence distance between the two planes will be $\sqrt{6}$

5. **Ans (A)**

$$b^2 - 4ac = p^2 - 4aq$$

$$\Rightarrow (b - p)(b + p) = 4a(c - q)$$

$$\alpha = \frac{q - c}{b - p}$$

Sum of roots of

$$f(x) = -\frac{b}{a} \text{ and that of } g(x) = -\frac{p}{a}$$

$$\text{AM of all roots} = -\frac{(b + p)}{4a} = \frac{q - c}{b - p} = \alpha$$

6. **Ans (C)**

Total number of ways in which 8 persons can be selected = ${}^{12}C_8$. Out of these 495 cases, there are 3 cases when all 8 are chosen from only two countries. So, 492 ways

7. **Ans (B)**

$$x + y + z = 6 \quad \dots (1)$$

$$x + 2y + 3z = 14 \quad \dots (2)$$

$$3*(2) - (1) \text{ will result in } 2x + 5y + 8z = 36$$

So, when $\alpha = 8$ and $\beta \neq 36$ we will have no solution

8. Ans (A)

Maximum value of $(\sin x + \cos x)^2$ is 2 and minimum value of $3 - 2 \tan x + \tan^2 x$ is 2

Hence, $\sin x + \cos x = \pm\sqrt{2}$ and $\tan x = 1$ So,

$$x = 2n\pi + \frac{\pi}{4} \Rightarrow x = \frac{\pi}{4}$$

9. Ans (B)

A is the event that at least one die does not show 5

B is the event that sum of two numbers showing on the two dices is 8

$$P(A) = \frac{35}{36} \text{ (only one case (5, 5) is not allowed)}$$

$$P(A \cap B) = \frac{5}{36}$$

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{1}{7}$$

10. Ans (B)

$(\sim(\sim r \wedge \sim p))$ is equivalent to $r \vee p$

And $(p \Rightarrow q)$ is equivalent to $\sim p \vee q$

$(p' \cup q) \cup (r \cup p)$ is universal set

11. Ans (D)

Number of terms in the expansion of

$$(2 - 3x + 4x^2)^n \text{ is } 2n + 1$$

So, $n = 6$

To find sum of all coefficients, put $x = 1$ to get 729

12. Ans (B)

Point of intersection of transverse tangents divides the line (internally) joining the centers in ratio of the radius of the circles

So, it divides $(12, -1)$ and $(-10, 3)$ in 5 : 15 or 1 : 3 ratio

13. Ans (A)

$$A = \begin{bmatrix} 3 & 2 \\ 1 & -4 \end{bmatrix}, \text{ adj } A = \begin{bmatrix} -4 & -2 \\ -1 & 3 \end{bmatrix}$$

$$A(\text{adj } A) = \begin{bmatrix} -14 & 0 \\ 0 & -14 \end{bmatrix} = -14I$$

14. Ans (C)

$$(\vec{a} + \mu \vec{b}) \cdot \vec{n} = 0 \Rightarrow \mu = -\frac{\vec{a} \cdot \vec{n}}{\vec{b} \cdot \vec{n}}$$

$$\vec{r} = \vec{a} - \left(\frac{\vec{a} \cdot \vec{n}}{\vec{b} \cdot \vec{n}} \right) \vec{b}$$

15. Ans (B)

All three sides of the triangle are equal

$$\text{So, } \sqrt{a^2 \sec^2 \theta + b^2 \tan^2 \theta} = 2a \sec \theta$$

$$b^2 \tan^2 \theta = 3a^2 \sec^2 \theta$$

$$\frac{b^2}{a^2} = 3 \cos^2 \theta > 3$$

$$e^2 = 1 + \frac{b^2}{a^2} > 4$$

$$e > 2$$

16. Ans (B)

$$2 \cos x = 3 \tan x$$

$$\Rightarrow 2 \cos^2 x = 3 \sin x$$

$$\Rightarrow 2 \sin^2 x + 3 \sin x - 2 = 0$$

$$\Rightarrow x = \frac{\pi}{6}$$

Area bounded by the curves and y axis will be

$$\begin{aligned} & \int_0^{\frac{\pi}{6}} (2 \cos x - 3 \tan x) dx \\ &= 2 \sin x - 3 \ln |\sec x| \Big|_0^{\frac{\pi}{6}} = 1 - 3 \ln \left(\frac{2}{\sqrt{3}} \right) \\ &= 1 - 3 \ln 2 + \frac{3}{2} \ln 3 \end{aligned}$$

17. Ans (B)

$p'''(x) = 0$ means $p(x)$ is a two - degree polynomial

Since $\lim_{x \rightarrow 2} f(x)$ exists, $p(x)$ will be divisible

by $x - 2$

$$\Rightarrow p(x) = a(x - 2)(x - b)$$

$$\text{So, } a(2 - b) = 7$$

$$\text{Since, } p(3) = 9, a(3 - b) = 9$$

Dividing them we will get $b = -\frac{3}{2}$ and $a = 2$

$$\text{So, } p(x) = 2x^2 - x - 6$$

18. **Ans (A)**

$$\frac{1}{y} \frac{dy}{dx} - \log y = e^x$$

Put $\log y = t$, $\frac{1}{y} \frac{dy}{dx} = \frac{dt}{dx}$

$$\frac{dt}{dx} - t = e^x$$

Integrating factor $= e^{-x}$

So, $te^{-x} = x + c$

$$t = xe^x + ce^x$$

$$y = e^{xe^x + ce^x}$$

Since, $f(0) = 1$, $c = 0$

$$f(x) = e^{xe^x} \Rightarrow f(1) = e^e$$

19. **Ans (B)**

$$A = \frac{a+b}{2}, G = \sqrt{ab} \text{ and } H = \frac{2ab}{a+b}$$

So, A, G and H are in GP

$$xA = yG = zH = k$$

$$x = \frac{k}{A}, y = \frac{k}{G}, z = \frac{k}{H}$$

Hence x, y, z are in GP

20. **Ans (D)**

$$f(x) = \begin{cases} \int_0^x (1 + |1-t|) dt, & x > 2 \\ 5x-7, & x \leq 2 \end{cases}$$

$$= \begin{cases} \int_0^1 (2-t) dt + \int_1^x t dt, & x > 2 \\ 5x-7, & x \leq 2 \end{cases}$$

$$= \begin{cases} 1 + \frac{x^2}{2}, & x > 2 \\ 5x-7, & x \leq 2 \end{cases}$$

So, clearly $f(x)$ is continuous at $x = 2$ but not differentiable

PART-3 : MATHEMATICS

1. **Ans (0.50)**

$$\cos \frac{99\pi}{7} - \cos \frac{54\pi}{7} - \cos \frac{60\pi}{7} =$$

$$\cos \frac{\pi}{7} + \cos \frac{5\pi}{7} + \cos \frac{3\pi}{7}$$

$$\frac{\sin \frac{3\pi}{7} \cos \frac{3\pi}{7}}{\sin \frac{\pi}{7}} = \frac{1}{2}$$

2. **Ans (1.5)**

$$\text{Let } y = \frac{3}{2-x^2} = f(x) \dots (1)$$

The function y is not defined for

$$x = \pm \sqrt{2}$$

$$\text{From (1), } x^2 = \frac{2y-3}{y}$$

$$\text{since for real } x, x^2 \geq 0,$$

$$\text{We have } \frac{2y-3}{y} \geq 0$$

$$y \geq 3/2 \text{ or } y < 0$$

Hence the range of the function is

$$(-\infty, 0) \cup \left[\frac{3}{2}, \infty \right)$$

3. **Ans (0.00)**

Only point where we need to check differentiability is 0

$$f(x) = \begin{cases} 4x^2(1-2x)^2, & x < 0 \\ 0, & x \geq 0 \end{cases}$$

$$f'(0) = \begin{cases} 0, & x < 0 \\ 0, & x \geq 0 \end{cases}$$

So, differentiable at $x = 0$

4. **Ans (17.00)**

$$z = x + iy$$

$$y = 8 \text{ and } x + \sqrt{x^2 + y^2} = 2$$

$$x + \sqrt{x^2 + 64} = 2$$

$$x^2 + 64 = x^2 - 4x + 4$$

$$x = -15$$

$$z = -15 + 8i, |z| = 17$$

5. **Ans (10.00)**

If width of river is 'd' and 'h' is height of the tree, then

$$d \tan 60^\circ = h$$

$$\text{and } (d+20) \tan 30^\circ = h$$

$$\text{So, } d = 10$$

6. Ans (8.00)

Limiting case will be when the line touches the parabola

So, discriminant of $(x + 2)^2 = ax$ should be 0

$$x^2 + (4 - a)x + 4 = 0$$

$$a^2 - 8a = 0 \Rightarrow a = 8$$

$$\Rightarrow \text{Length of latus rectum} = 8$$

7. Ans (35.00)

Total burning time = $30 + 40 + 50 = 120$

If t is time for which exactly two candles are burning, then

$$10 \cdot 3 + 20 + 2t = 120$$

$$\text{So, } t = 35$$

8. Ans (3.00)

If number arranged in ascending order are $x, 1,$

$$4, 10, 12, \text{ then } \frac{27+x}{5} = 4 \Rightarrow x = -7$$

If number arranged in ascending order are $1, x,$

$4, 10, 12,$ then also $x = -7$ which is not possible

If number arranged in ascending order are $1, 4,$

$$x, 10, 12, \text{ then } \frac{27+x}{5} = x \Rightarrow x = 6.75$$

If number arranged in ascending order are $1, 4,$

$$10, x, 12, \text{ then } \frac{27+x}{5} = 10 \Rightarrow x = 23$$

which is not possible

If number arranged in ascending order are $1, 4,$

$$10, 12, x \text{ then } \frac{27+x}{5} = 10 \Rightarrow x = 23$$

So, three possibilities

9. Ans (4.00)

If $y - 2x = k$, then k will take its extreme values when this straight line is tangent to the given ellipse

Equation of tangent to the ellipse and having slope 2 is

$$y = 2x \pm \sqrt{3 \cdot 4 + 4} \Rightarrow y = 2x \pm 4$$

$$\text{So, } k \in [-4, 4]$$

10. Ans (2.00)

$$\int_{-\pi}^{\pi} \frac{2x(1 + \sin x)}{1 + \cos^2 x} dx$$

$$= \int_{-\pi}^{\pi} \frac{2x}{1 + \cos^2 x} dx + \int_{-\pi}^{\pi} \frac{2x \sin x}{1 + \cos^2 x} dx$$

$\frac{2x}{1 + \cos^2 x}$ is an odd function and $\frac{2x \sin x}{1 + \cos^2 x}$ is an even function

$$I = \int_{-\pi}^{\pi} \frac{2x(1 + \sin x)}{1 + \cos^2 x} dx = 0 + 2 \int_0^{\pi} \frac{2x \sin x}{1 + \cos^2 x} dx$$

$$I = 2 \int_0^{\pi} \frac{2(\pi - x) \sin x}{1 + \cos^2 x} dx$$

$$\text{Add them to get } I = \pi \int_0^{\pi} \frac{2 \sin x}{1 + \cos^2 x} dx$$

$$\text{Put } \cos x = t, -\sin x dx = dt$$

$$I = \pi \int_{-1}^1 \frac{2}{1 + t^2} dt = \pi^2$$